KAIZEN MATHS

Lesson starter

Algebraic Fractions

Adding and Subtracting

Name _____

S1

Simplify:

$\frac{3x-4}{(x+6)} + \frac{4x+4}{(x+6)}$

S2

Simplify:

$\frac{3}{x} + \frac{2}{(x+4)}$

S3

Simplify:

$\frac{2x-6}{(x+6)} + \frac{3x-5}{(x+6)}$

S4

Simplify:

$\frac{5}{(x+3)} + \frac{2}{(x-5)}$

Show your working

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S4

Simplify:

$\frac{5}{(x+3)} + \frac{2}{(x-5)}$

ALG-49C4-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Algebraic Fractions

Adding and Subtracting

Name _____

E1

Simplify:

$\frac{3x-5}{x} + \frac{x+5}{x}$

E2

Simplify:

$\frac{2}{(x-5)} + \frac{5}{(x+4)}$

E3

Simplify:

$\frac{5x-5}{(x-6)} + \frac{5x+2}{(x-6)}$

KAIZEN MATHS

Exit ticket

Algebraic Fractions

Adding and Subtracting

Name _____

E1

Simplify:

$\frac{3x-5}{x} + \frac{x+5}{x}$

E2

Simplify:

$\frac{2}{(x-5)} + \frac{5}{(x+4)}$

E3

Simplify:

$\frac{5x-5}{(x-6)} + \frac{5x+2}{(x-6)}$

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Show your working

KAIZEN MATHS

Exit ticket

Algebraic Fractions

Adding and Subtracting

Name _____

E1

Simplify:

$\frac{3x-5}{x} + \frac{x+5}{x}$

E2

Simplify:

$\frac{2}{(x-5)} + \frac{5}{(x+4)}$

E3

Simplify:

$\frac{5x-5}{(x-6)} + \frac{5x+2}{(x-6)}$

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Show your working

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Exit ticket

Algebraic Fractions

Adding and Subtracting

Name _____

E1

Simplify:

$\frac{3x-5}{x} + \frac{x+5}{x}$

E2

Simplify:

$\frac{2}{(x-5)} + \frac{5}{(x+4)}$

E3

Simplify:

$\frac{5x-5}{(x-6)} + \frac{5x+2}{(x-6)}$

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Show your working

Teacher answers & working

Algebraic Fractions · Adding and Subtracting

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S1

Simplify:

$\frac{3x-4}{(x+6)} + \frac{4x+4}{(x+6)}$

Answer:

$\frac{7x}{(x+6)}$

Working:

Step 1: The denominators are the same, so keep the denominator and add the numerators.

$\frac{3x-4}{(x+6)} + \frac{4x+4}{(x+6)} = \frac{(3x-4)+(4x+4)}{(x+6)}$

Step 2: Collect like terms in the numerator.

$(3x - 4) + (4x + 4) = 7x$

Step 3: Write the combined numerator over the original denominator.

$\frac{7x}{(x+6)}$

Restriction: x is not equal to -6 because the denominator cannot be zero.

S2

Simplify:

$\frac{3}{x} + \frac{2}{(x+4)}$

Answer:

$\frac{5x+12}{x(x+4)}$

Working:

Step 1: The denominators are different, so use the common denominator x(x + 4).

Step 2: Multiply the first fraction by the missing factor (x + 4).

$\frac{3}{x} = \frac{3(x+4)}{x(x+4)}$

Step 3: Multiply the second fraction by the missing factor x.

$\frac{2}{(x+4)} = \frac{2x}{x(x+4)}$

Step 4: Add the numerators over the common denominator.

$\frac{3(x+4)+2x}{x(x+4)}$

Step 5: Expand and collect the numerator to get 5x + 12.

Step 6: The simplified fraction is $\frac{5x+12}{x(x+4)}$.

Teacher answers & working

Algebraic Fractions · Adding and Subtracting

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S3

Simplify:

$\frac{2x-6}{(x+6)} + \frac{3x-5}{(x+6)}$

Answer:

$\frac{5x-11}{(x+6)}$

Working:

Step 1: The denominators are the same, so keep the denominator and add the numerators.

$\frac{2x-6}{(x+6)} + \frac{3x-5}{(x+6)} = \frac{(2x-6)+(3x-5)}{(x+6)}$

Step 2: Collect like terms in the numerator.

$(2x - 6) + (3x - 5) = 5x - 11$

Step 3: Write the combined numerator over the original denominator.

$\frac{5x-11}{(x+6)}$

Restriction: x is not equal to -6 because the denominator cannot be zero.

S4

Simplify:

$\frac{5}{(x+3)} + \frac{2}{(x-5)}$

Answer:

$\frac{7x-19}{(x+3)(x-5)}$

Working:

Step 1: The denominators are different, so use the common denominator (x + 3)(x - 5).

Step 2: Multiply the first fraction by the missing factor (x - 5).

$\frac{5}{(x+3)} = \frac{5(x-5)}{(x+3)(x-5)}$

Step 3: Multiply the second fraction by the missing factor (x + 3).

$\frac{2}{(x-5)} = \frac{2(x+3)}{(x+3)(x-5)}$

Step 4: Add the numerators over the common denominator.

$\frac{5(x-5)+2(x+3)}{(x+3)(x-5)}$

Step 5: Expand and collect the numerator to get 7x - 19.

Step 6: The simplified fraction is $\frac{7x-19}{(x+3)(x-5)}$.

Teacher answers & working

Algebraic Fractions · Adding and Subtracting

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E1

Simplify:

$\frac{3x-5}{x} + \frac{x+5}{x}$

Answer:

$\frac{4x}{x}$

Working:

Step 1: The denominators are the same, so keep the denominator and add the numerators.

$\frac{3x-5}{x} + \frac{x+5}{x} = \frac{(3x-5)+(x+5)}{x}$

Step 2: Collect like terms in the numerator.

$(3x - 5) + (x + 5) = 4x$

Step 3: Write the combined numerator over the original denominator.

$\frac{4x}{x}$

Restriction: x is not equal to 0 because the denominator cannot be zero.

E2

Simplify:

$\frac{2}{(x-5)} + \frac{5}{(x+4)}$

Answer:

$\frac{7x-17}{(x-5)(x+4)}$

Working:

Step 1: The denominators are different, so use the common denominator (x - 5)(x + 4).

Step 2: Multiply the first fraction by the missing factor (x + 4).

$\frac{2}{(x-5)} = \frac{2(x+4)}{(x-5)(x+4)}$

Step 3: Multiply the second fraction by the missing factor (x - 5).

$\frac{5}{(x+4)} = \frac{5(x-5)}{(x-5)(x+4)}$

Step 4: Add the numerators over the common denominator.

$\frac{2(x+4)+5(x-5)}{(x-5)(x+4)}$

Step 5: Expand and collect the numerator to get 7x - 17.

Step 6: The simplified fraction is $\frac{7x-17}{(x-5)(x+4)}$.

Teacher answers & working

Algebraic Fractions · Adding and Subtracting

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E3

Simplify:

$\frac{5x-5}{(x-6)} + \frac{5x+2}{(x-6)}$

Answer:

$\frac{10x-3}{(x-6)}$

Working:

Step 1: The denominators are the same, so keep the denominator and add the numerators.

$\frac{5x-5}{(x-6)} + \frac{5x+2}{(x-6)} = \frac{(5x-5)+(5x+2)}{(x-6)}$

Step 2: Collect like terms in the numerator.

$(5x - 5) + (5x + 2) = 10x - 3$

Step 3: Write the combined numerator over the original denominator.

$\frac{10x-3}{(x-6)}$

Restriction: x is not equal to 6 because the denominator cannot be zero.